

Analysis 1: Aug-Nov 2026

Exercises (31 Aug 2026).

Part 1

Throughout this set of exercises, $\langle a_n : n \geq 1 \rangle$, $\langle b_n : n \geq 1 \rangle$, $\langle c_n : n \geq 1 \rangle$, $\langle d_n : n \geq 1 \rangle$, represent sequences of real numbers. In each of the following questions, prove if statement is true and give a counter example if not true.

- (a) Suppose $|a_n| \leq 10^9$ for all $n \geq 1$. Then $\exists$ subsequence $\langle a_{n_k} : k \geq 1 \rangle$ of $\langle a_n : n \geq 1 \rangle$, such that $\langle a_{n_k} : k \geq 1 \rangle$ converges.
- (b) If $-\infty < \liminf_n a_n \leq \limsup_n a_n < \infty$. Then the sequence $\langle a_n : n \geq 1 \rangle$ is bounded.
- (c) If $-\infty < \liminf_n a_n = \limsup_n a_n < \infty$. Then the sequence $\langle a_n : n \geq 1 \rangle$ is convergent.
- (d) Let $a_n > 0$ for all n and let $a_n^* = \max\{a_j : 1 \leq j \leq n\}$, $b_n = \frac{a_n}{a_n^*}$. If $\langle b_n : n \geq 1 \rangle$ converges, we can conclude that $\langle a_n : n \geq 1 \rangle$ is convergent.
- (e) Let $a_n > 0$ for all n . For $m \geq 1$ let $b_m = 2^m a_{2^m}$. If the series $\sum_m b_m$ is convergent, then the series $\sum_n a_n$ is also convergent. (if true, prove from first principles).
- (f) Let $s_n = \sum_{k=1}^n a_k$ and $t_n = \sum_{k=1}^n a_k^5$. If $\langle s_n : n \geq 1 \rangle$ converges (to a finite number $s \in \mathbb{R}$), then $\langle t_n : n \geq 1 \rangle$ also converges.
- (g) Suppose $a_n \leq b_n \leq c_n$ and $u_n = \sum_{k=1}^n a_k$ and $t_n = \sum_{k=1}^n b_k$ and $v_n = \sum_{k=1}^n c_k$. If $\langle u_n : n \geq 1 \rangle$ and $\langle v_n : n \geq 1 \rangle$ converge in $\mathbb{R}$ then the series $\sum_n b_n$ converges absolutely.

Part 2

Throughout this set of exercises, f, g, f_n (for $n \geq 1$) are continuous functions from $A \subseteq \mathbb{R}$ to $B \subseteq \mathbb{R}$. In each of the following questions, prove if statement is true and give a counter example if not true.

- (a) Suppose $A = [a, b]$ for some $a, b \in \mathbb{R}$. Then $\exists c, d \in \mathbb{R}$ such that $B \subseteq [c, d]$.
- (b) Suppose $A \subseteq (a, b)$ for some $a, b \in \mathbb{R}$. Then $\exists c, d \in \mathbb{R}$ such that $B \subseteq (c, d)$.
- (c) Suppose $A = [a, b]$ for some $a, b \in \mathbb{R}$. Then $\exists u, v \in A$ such that $f(u) \leq f(x) \leq f(v)$, $\forall x \in A$.
- (d) Suppose $A = [a, b]$ for some $a, b \in \mathbb{R}$ with $f(a) > 0$ and $f(b) < 0$. Let $u_0 = 1$ and $v_0 = b$. We will define u_n, v_n inductively as follows. For $n \geq 0$ let $c_{n+1} = \frac{1}{2}(u_n + v_n)$. If $f(c_{n+1}) \geq 0$, then $u_{n+1} = c_{n+1}$ and $v_{n+1} = v_n$ and if $f(c_{n+1}) < 0$, then $u_{n+1} = u_n$ and $v_{n+1} = c_{n+1}$. Then
 - (i) $f(u_n) \geq 0$ and $f(v_n) < 0$ for all n .
 - (ii) $|v_n - u_n| \leq 2^{-n}|b - a|$
 - (iii) $\langle u_n : n \geq 1 \rangle$ converges to u^* such that $f(u^*) = 0$.
- (e) Suppose $B \subseteq (0, \infty)$. Let $h(x) = \frac{f(x)}{g(x)}$ for $x \in A$. Then h is continuous.
- (f) Suppose $B = A$ and let $h(x) = g(f(x))$ for $x \in A$. Then h is continuous.
- (g) Let $A = [0, 10]$. Show that f is uniformly continuous, i.e. $\forall \epsilon > 0 \exists \delta > 0$ such that $x, y \in A$, $|x - y| < \delta$ implies $|f(x) - f(y)| < \epsilon$.
- (h) Let $A = [-10, 10]$ and for $x \in A$ and $n \geq 1$, let $f_n(x) = \sum_{k=1}^n \frac{x^k}{n!}$. Then for all $x \in A$, $f_n(x)$ converges to (say) $f(x)$ and further $\forall \epsilon > 0, \exists n^*$ such that $|f_n(x) - f(x)| \leq \epsilon$ for all $x \in A$ and $n \geq n^*$.